

Open Science Research Integrity Toolkit

27 empirical techniques for advancing research integrity – organized by the level of the problem (tier) and issue type

TIER 1

Individual-Level Misconduct & Errors

Tools that examine a single paper's statistics, images, or text for signs of fabrication, manipulation, or error — without requiring access to raw data.

Data Fabrication

A researcher invents or alters data. These tools detect impossible or implausible summary statistics in published papers.

- | | | |
|----|------------------|---|
| 01 | GRIM Test | Tests whether a reported mean is possible given sample size and integer scale. Requires only the reported mean and N — no raw data. |
| 02 | SPRITE | Reconstructs all integer datasets consistent with reported M, SD, and N. Exposes physically impossible or implausible data distributions. |

Image Manipulation

Figures are duplicated, spliced, or AI-generated. Tools identify pixel-level duplication patterns within or across manuscripts.

- | | | |
|----|-----------------------------|---|
| 03 | Bik Visual Screening | Manual comparison of published figures for identical regions passed off as distinct experiments. Requires full-resolution PDF or image files. |
| 04 | Imagetwin AI | AI-powered batch scanning for pixel-level image duplication. Deployed by publishers at submission. Requires good-quality PDF input. |
| 05 | Proofig AI | Publisher-grade image integrity checking before peer review. Detects duplicated panels, background uniformity, and contrast manipulation. |

Plagiarism & Text Recycling

Text is copied or AI-generated without attribution. Detection is harder as paraphraser preserve meaning while evading exact-match tools.

- | | | |
|----|----------------------------------|--|
| 06 | iThenticate / CrossCheck | Text similarity matching against published papers, preprints, and web content. Cannot detect meaning-preserving paraphrase. |
| 07 | Tortured Phrases Screener | Flags machine-translation obfuscation used to evade plagiarism detectors — signature of paper-mill produced manuscripts. |
| 08 | AI-Text Detectors | Identifies probabilistic signatures of LLM-generated text. Reliability is limited after deliberate paraphrasing; use as a screening flag only. |

Research Methods Errors

Coding bugs and data-entry errors that are unintentional but consequential. Tools reconstruct implied computations and flag implausible values.

- | | | |
|----|------------------|--|
| 09 | statcheck | Extracts t, F, χ^2 , z, and r statistics from a PDF and recalculates p-values. Flags decision-changing mismatches. Requires machine-readable PDF. |
|----|------------------|--|

TIER 2a

Collaborative Pipeline Failure

Errors emerging from handoffs between team members on interdependent code and data. One upstream change cascades silently through all downstream components.

- | | | |
|----|---|--|
| 10 | Registered vs. Published Outcome Audit | Compares pre-registered hypotheses and outcomes to those in the final paper. Divergences signal HARKing, outcome switching, or pipeline changes. |
|----|---|--|

Team-Level Analytic Issues

Tools that detect problems arising from collective analytic decisions — p-hacking, selective reporting, model fragility — applied across a set of studies or specifications.

p-Hacking / Selective Reporting

Teams run many analyses and report only significant results. Tools examine p-value distributions across a literature to detect selective reporting signatures.

11	p-curve	Plots distribution of significant p-values. Genuine effects → right-skewed. p-hacking → flat or left-skewed. Requires test statistics and df per finding.
12	p-uniform	Models full p-value distribution under a specific assumed effect to estimate the true effect size and test publication bias. Requires effect sizes and p-values.
13	Caliper Test	Tests for excess of results just below $p < .05$ — a signature of selective stopping or rounding. Requires p-values from many studies in the same literature.
14	z-curve 2.0	Fits a mixture model to all z-scores (significant and non-significant). Estimates expected replication rate and discovery rate. Requires 100+ test statistics.

Fragile Findings & Analytical Fragility

Results that depend on undisclosed analytic choices. Tools run all defensible specifications to reveal whether significance is robust or narrow.

The Garden of Forking Paths

Three tools for mapping how much a finding depends on the particular choices made from a larger space of equally defensible analytic options.

15	Multiverse Analysis	Runs every defensible combination of analytic choices and plots the distribution of resulting effect sizes. Requires access to raw data.
16	Specification Curve Analysis	Adds inferential permutation tests to the multiverse approach — asks whether observed results exceed a null baseline. Requires raw data.
17	Vibration of Effects	Cycles through all subsets of candidate covariates; records effect estimate for each. VoE ratio > 2 or sign changes indicate fragility. Requires individual-level data.

Publication Bias (Team & PI Decisions)

Teams selectively submit significant results. Tools detect the statistical signature of missing studies and estimate inflation of the published effect size.

Correcting for the File Drawer

Four meta-analytic tools that detect and correct for publication bias — the systematic suppression of null results from the published record.

18	Funnel Plot & Egger Test	Plots effect size vs. precision. Asymmetry signals missing null studies. Egger test formalizes this as a regression. Requires ≥ 10 studies, effect sizes and SEs.
19	Trim-and-Fill	Imputes missing studies to restore funnel symmetry; re-estimates the bias-corrected pooled effect. Assumes missing studies mirror observed asymmetric ones.
20	PET-PEESE	Meta-regression: regresses effect size on SE (PET) then SE^2 (PEESE) to estimate the true effect after removing bias. Best with ≥ 20 –30 studies.
21	Excess Significance Test (TES)	Compares observed to expected number of significant results given study power. Large excess implies p-hacking, publication bias, or both.

TIER 2b

Group / Collusion-Level Misconduct

Tools that detect coordinated misconduct across multiple authors, institutions, or journals — analyzing relationships between entities rather than individual paper content.

Citation Rings & Uncritical Peer Review

Dense mutual-citation or mutual-review networks inconsistent with field norms. Detection requires bibliometric analysis of relational patterns.

- | | | |
|-----------|---------------------------------------|--|
| 22 | Citation Network Analysis | Maps citation relationships to identify anomalously dense clusters. Uses network centrality and modularity. Requires access to bibliometric databases. |
| 23 | Bibliometric Anomaly Detection | Flags journals or author groups with anomalous self-citation or co-author citation rates compared to field baselines. Requires citation metadata. |

Fake Peer Review Rings

Authors submit fabricated reviewer emails or collude with editors for pre-agreed reviews. Now accounts for roughly half of all retractions globally.

- | | | |
|-----------|--------------------------------|---|
| 24 | Editor Network Analysis | Maps editorial submission routing to identify editors disproportionately handling each other's manuscripts. Requires publisher submission metadata. |
|-----------|--------------------------------|---|

Paper Mills & Predatory Journals

Industrial fabrication networks producing manuscripts for purchase. Detection uses text-level obfuscation signatures and image clustering across large paper sets.

- | | | |
|-----------|-----------------------------------|---|
| 25 | Problematic Paper Screener | Scans papers for tortured phrases, image duplication, and metadata anomalies. Has flagged 32,786+ suspected paper-mill products. Works from PDF and metadata. |
| 26 | Hijacked Journal Checker | Database of 400+ journal ISSNs cloned by scam publishers mimicking legitimate journals. Functions as a lookup tool — requires only journal name or ISSN. |

TIER 3

Field / Community-Level Problems

Tools that assess the integrity or reliability of an entire research literature — quantifying systemic biases in how knowledge is produced, reported, and preserved.

Poor Data Stewardship & Inaccessible Research

Data and code are lost, unavailable despite stated commitments, or deposited in unusable formats. Tools assess whether research meets minimum reproducibility standards.

- | | | |
|-----------|---|--|
| 27 | FAIR Compliance Assessment Tools | Automated scoring (FAIRshake, F-UJI) of whether a dataset meets the FAIR principles: Findable, Accessible, Interoperable, Reusable. Requires only the repository landing page. |
|-----------|---|--|